EXPLORING AND PREDICTING BREAST CANCER MALIGNANCY USING SIZE AND MORPHOLOGY FEATURES

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INTRODUCTION – BREAST CANCER

- Most common cancer diagnosis for women worldwide, according to the CDC (Esen et al., 2024).
- According to the National Cancer Institute of the United States 13.4% of women born today will receive a breast cancer diagnosis during in their lifetime (Vig, 2014).
- Diagnosis currently relies on imaging and fine-needle aspiration to assess tumor characteristics (Esen et al., 2024).
- Countries with a low Human Development Index have fewer diagnoses and more deaths (WHO, 2023).

LITERATURE REVIEW

Tumor Size & Morphology as Predictors

- Bigger tumors means higher nodal spread or mortality but size alone is insufficient (Sopik, 2018; Min, 2021).
- Ultrasound stiffness + morphology improves benign/malignant separation (Ibrahim, 2023).
- On WDBC, size metrics (radius/area/ perimeter) are strong but better with appearance features (Arravalli, 2025).
- EFA: a dominant size factor exists with additional appearance factors explaining variance (Kumar, 2020).

PCA for Dimensionality Reduction

- PCA reduces redundancy and boosts accuracy; PCA+LDA > raw LDA on WDBC (Esen, 2024).
- Kernel-PCA preserves nonlinear class structure → higher accuracy (Mushtaq, 2023).
- Info Gain + PCA ranks & compresses features → best overall performance (Li, 2023).
- PCA tackles collinearity and stabilizes classifiers (Massafra, 2021).
- Preprocessing PCA curbs overfitting and improves diverse ML models (Manikandan, 2023).

LITERATURE REVIEW

Linking Size & Appearance via CCA

- High canonical correlations between morphology and texture; fusion raises accuracy (Liu, 2022).
- KCCA captures nonlinear size, appearance links leading to better classification (Jabeen, 2024).
- Parallel/multi-view CCA aligns modalities for improved retrieval (Abderrahim, 2024).
- DGCCA integrates FFDM + DBT revealing strong latent cross-modal factors (Gupta, 2025).
- sCCA in multiomics finds sparse yet informative associations (Qi, 2021).

FNA/WDBC Evidence

- WDBC built from FNA nuclear descriptors (Street, 1993).
- Concavity & concave points are highly predictive (Street, 1993; Wolberg, 1990).
- Mathematical programming achieves high accuracy with compactness/symmetry (Mangasarian, 1995).
- Feature selection repeatedly surfaces concavity, area, compactness as top signals (Akben, 2019).
- Fuzzy-genetic approaches add robustness on small samples (Pena-Reyes, 1999).

RESEARCH QUESTIONS

1. Can PCA effectively reduce dimensionality while preserving variance?
2. Is tumor size the primary indicator of malignancy? (EFA, CA)
3. What is the relationship between size and appearance variables (CCA)?
4. Which tumor features are most predictive of malignancy (LASSO)?
5. How well can predictive models classify tumors (LDA, LASSO)?

DATASET

Breast Cancer Wisconsin (Diagnostic) Dataset -Medical diagnostics for breast cancer detection

- **Dataset Details**:
    - **569 patient records** (samples)
    - **32 columns**: 1 sample ID (to be dropped), 1 diagnosis label, 30 numeric features
- **Key Variables**:
    - **Numeric Features**: 30 continuous measurements (e.g., radius, texture, smoothness of cell nuclei)
    - **Categorical Variable**: 1 diagnosis label (Benign or Malignant)
- **No missing values**

METHODS

Exploratory Methods

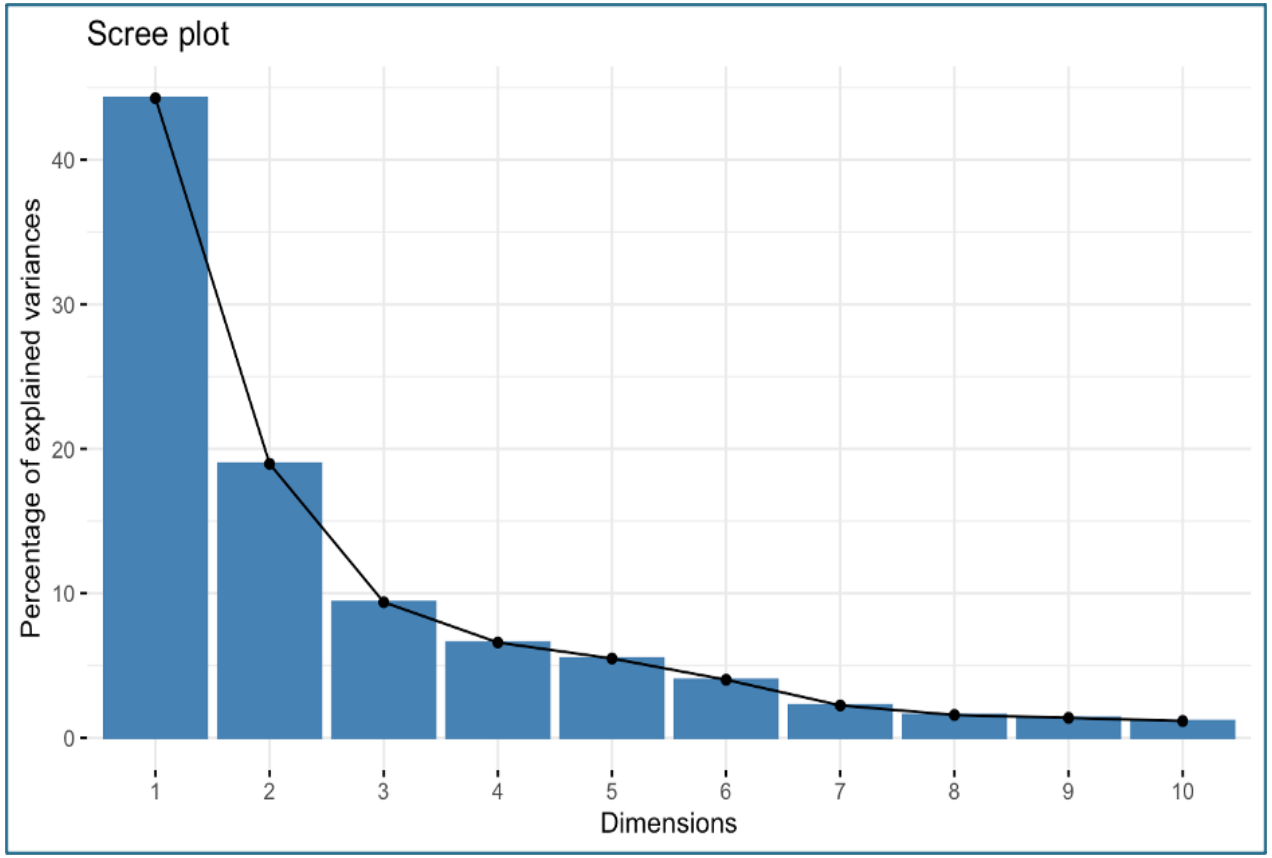
- Principal Component Analysis (PCA)
- Exploratory Factor Analysis (EFA)
  - Varimax Rotation
- Canonical Correlation Analysis (CCA)
- Correspondence Analysis(CA)

Classifying and Predictive Methods

- LASSO Regularized Regression
- Linear Discriminant Analysis (LDA)

PCA RESULTS/DISCUSSION

- Dimensionality reduction on 30 features
- 3 PCs → 70% variance explained
- 5 PCs → 80% variance explained
- 6 PCs → 89% variance explained
  - (eigenvalues > 1.0)



Scree plot of principal components showing variance explained

EXPLORATORY FACTOR ANALYSIS

- Two factors selected from the PCA
- Varimax rotation performed
- Size Factor - dominant
  - Radius, perimeter, area, etc.
- Formation factor - secondary
  - Smoothness, compactness, symmetry, etc.
- Accounting for 53% of the variance

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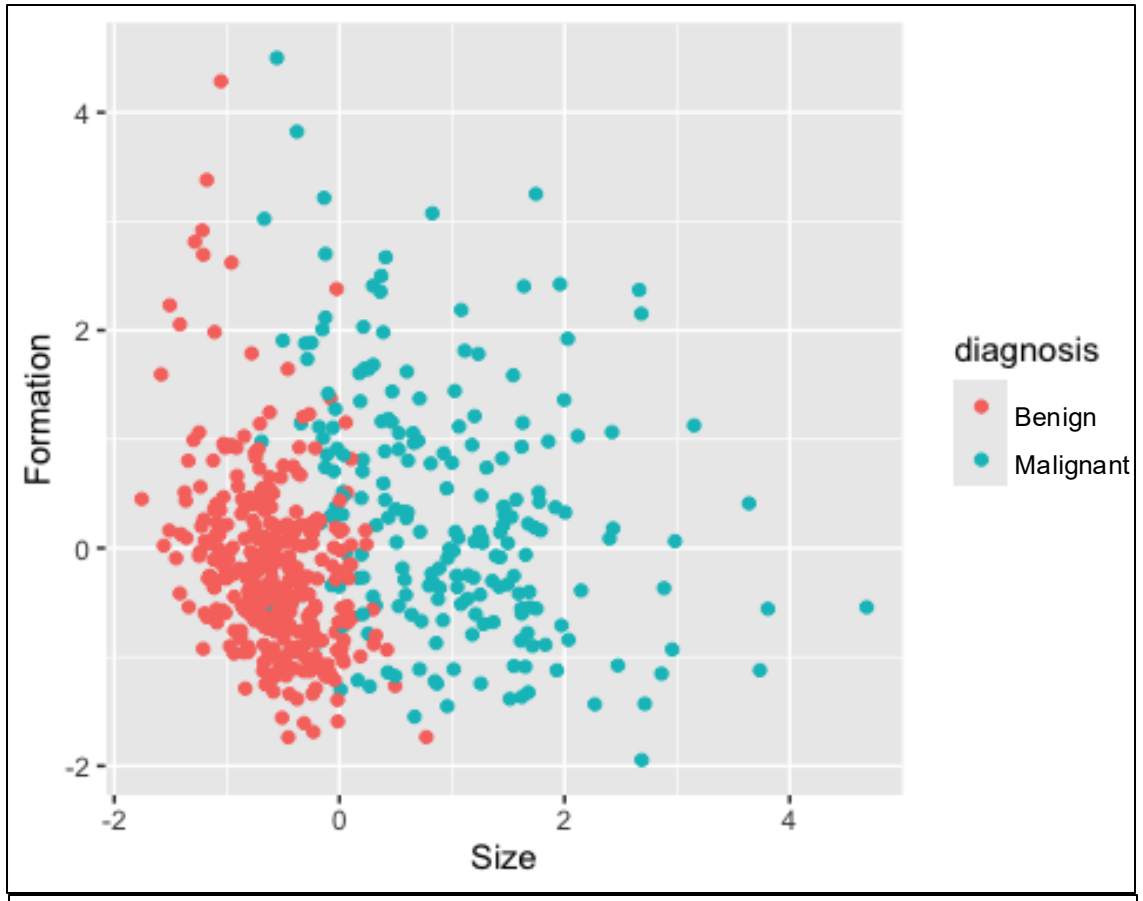
Loadings:
                        Factor1 Factor2
radius_mean             0.985
perimeter_mean          0.986
area_mean               0.984
concavity_mean          0.714
concave points_mean     0.851
radius_se               0.732
perimeter_se            0.725
area_se                 0.784
radius_worst            0.987
perimeter_worst         0.985
area_worst              0.973
concave points_worst    0.780
smoothness_mean                 0.641
compactness_mean                0.796
symmetry_mean                   0.602
fractal_dimension_mean          0.854
compactness_se                  0.757
concavity_se                    0.646
concave points_se               0.565
fractal_dimension_se            0.687
smoothness_worst                0.594
compactness_worst               0.740
concavity_worst                 0.675
symmetry_worst                  0.504
fractal_dimension_worst         0.829
texture_mean            0.349
texture_se                      0.138
smoothness_se                   0.330
symmetry_se                     0.344
texture_worst           0.334

               Factor1 Factor2
SS loadings      9.552   6.461
Proportion Var   0.318   0.215
Cumulative Var   0.318   0.534
```

Factor loadings calculated after the varimax rotation

EXPLORATORY FACTOR ANALYSIS

- Plotting Size vs Formation shows clear delineation along Size axis
- Supports size being more influential in malignancy results



Scatterplot comparing factors by diagnosis

RESULTS/DISCUSSION - CCA

- 2 main variable groups identified
- Size Variables:
  - Area, perimeter, radius, compactness
- Appearance Variables:
  - Texture, smoothness, compactness, concavity, concave points, symmetry, fractal dimension.
- 3 Significant Canonical Variates explain a large percentage of shared variance.

Canonical Correlations:			
CV1	CV2	CV3	CV4
0.96169	0.92798	0.26377	0.09034
Shared Variance on Each Canonical Variate:			
CV1	CV2	CV3	CV4
0.924845	0.861154	0.069576	0.008161

Bartlett's Chi-Squared Test:			
	rho^2	Chi-sq	Pr(>X)
CV 1	0.92484	2606.98	< 0.0001 ***
CV2	0.86115	1153.71	< 0.0001 ***
CV3	0.06958	45.09	< 0.0001 ***
CV4	0.00816	4.6	0.2

RESULTS/DISCUSSION - CCA

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Canonical Variate Coefficients:

        X Vars:
                        CV 1        CV 2        CV 3        CV 4
texture            -0.0070928   0.0040995   0.020084    0.05102
smoothness          5.2244598  12.1336331  33.758815   42.18538
concavity          -2.7319019   4.3065691 -30.531493   14.84420
concave points    -20.7823430 -17.4791179  50.036062  -45.59469
symmetry           -0.2946706   4.2769270   6.726946   36.50624
fractal_dimension -10.1423961 113.0571699  26.026491 -151.75738

        Y Vars:
                  CV 1       CV 2      CV 3      CV 4
radius       0.5241463  0.5527584 10.678253 -1.043819
perimeter   -0.0786427 -0.1788050 -1.624907  0.429489
area        -0.0014249  0.0034358  0.001361 -0.018705
compactness -9.8939369 23.3774309 51.940070 -9.800674
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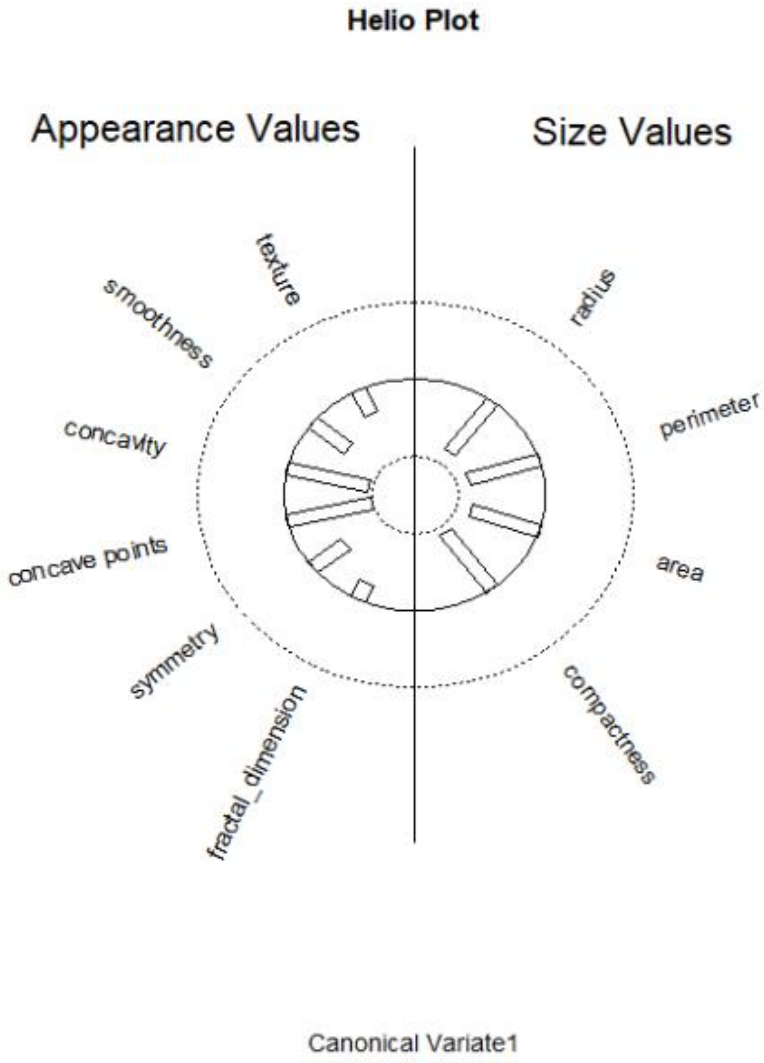
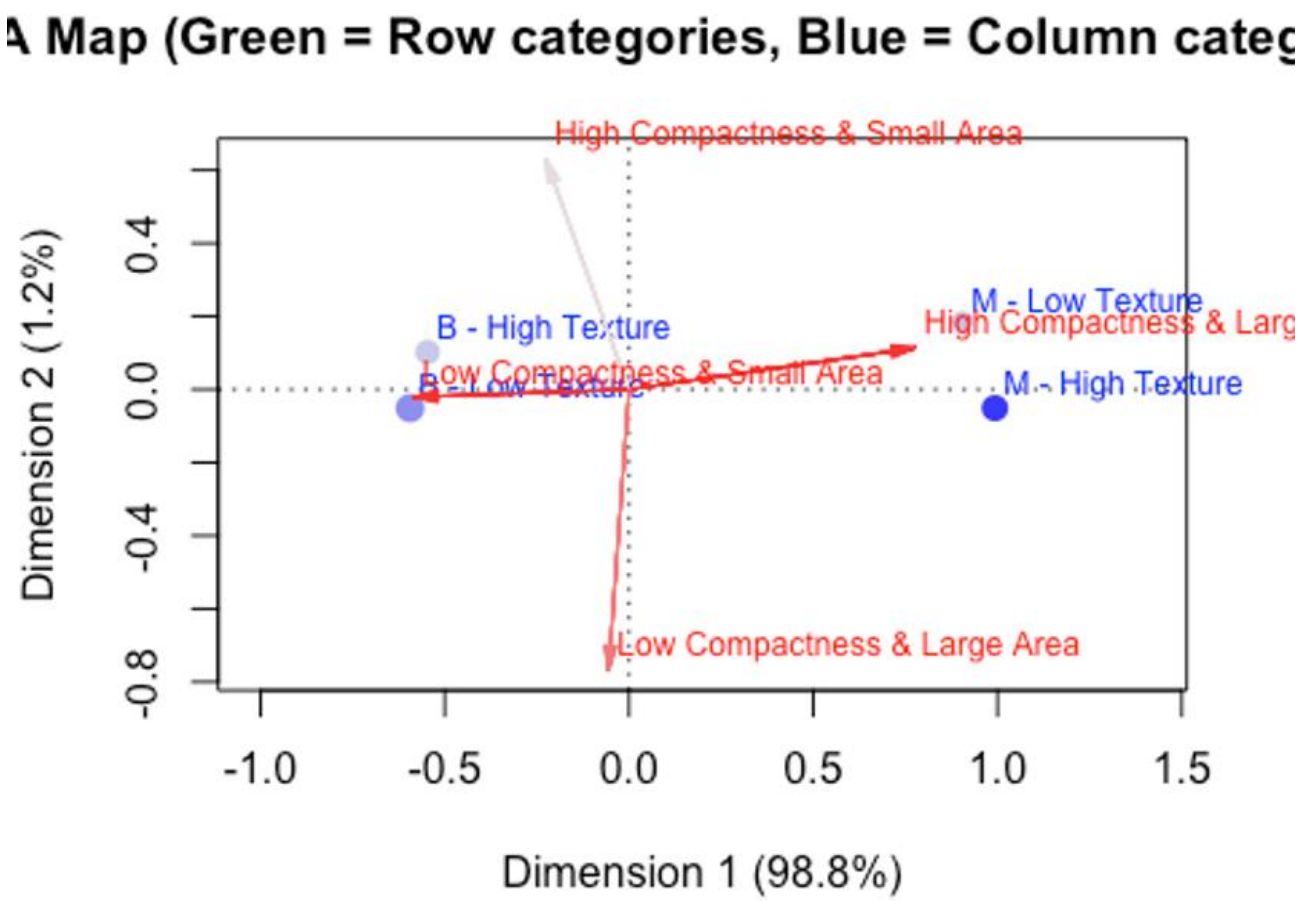
- CV1 dominated by 'concave points' and 'compactness' (perimeter^2/area).



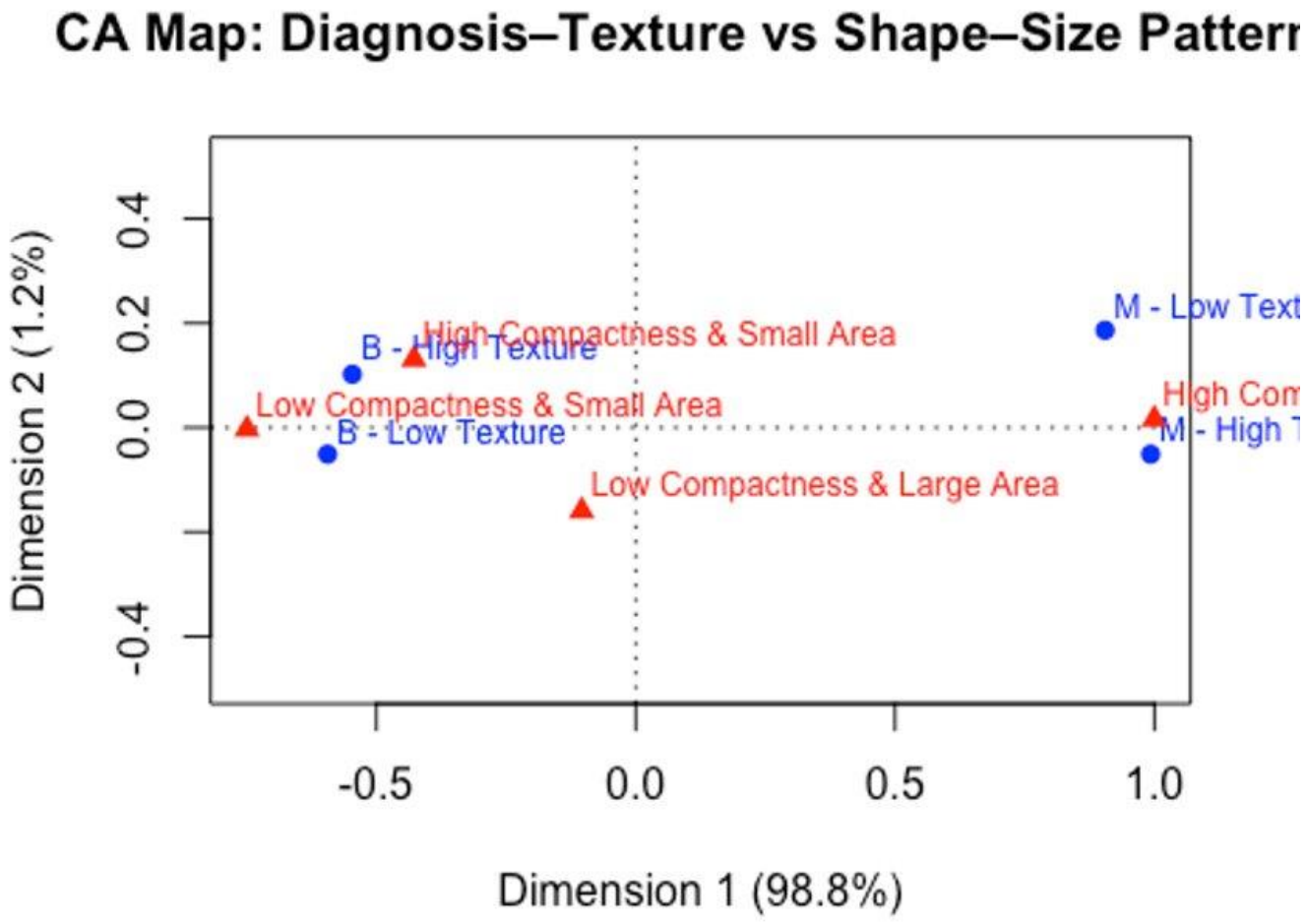
RESULTS/DISCUSSION
CORRESPONDENCE ANALYSIS

- **Combined compactness and area groupings -** Formed new column categories like "Compactness Group & Area Group" (e.g., High Compactness & Large Area)
- **Created texture-based row categories –** Grouped texture_mean into **Low Texture** and **High Texture** using the median split.

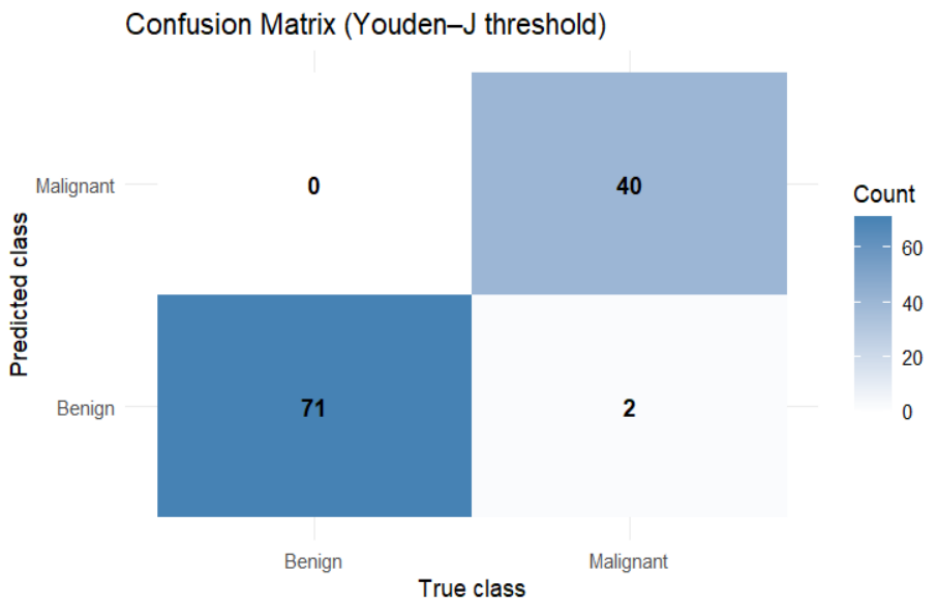


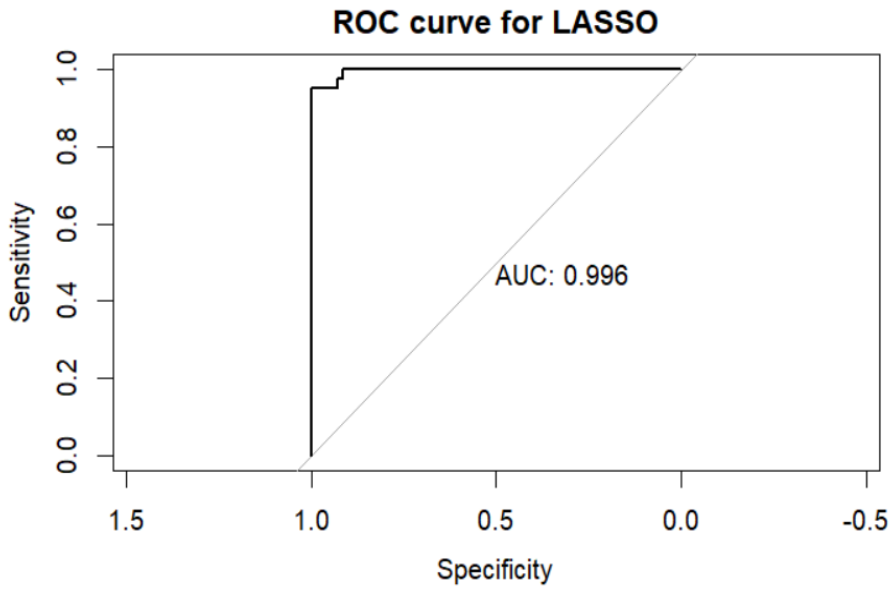
CORRESPONDENCE ANALYSIS TEXTURE VS SHAPE

- **Combined diagnosis with texture group -** Formed new row categories like "Diagnosis - Texture Group" (e.g., M - High Texture).
- **Created area-based column categories -** Grouped area_mean into **Small Area** and **Large Area** using the median.
- **Combined compactness and area groupings -** Formed new column categories like "Compactness Group & Area Group" (e.g., High Compactness & Large Area).



LASSO RESULTS/DISCUSSION





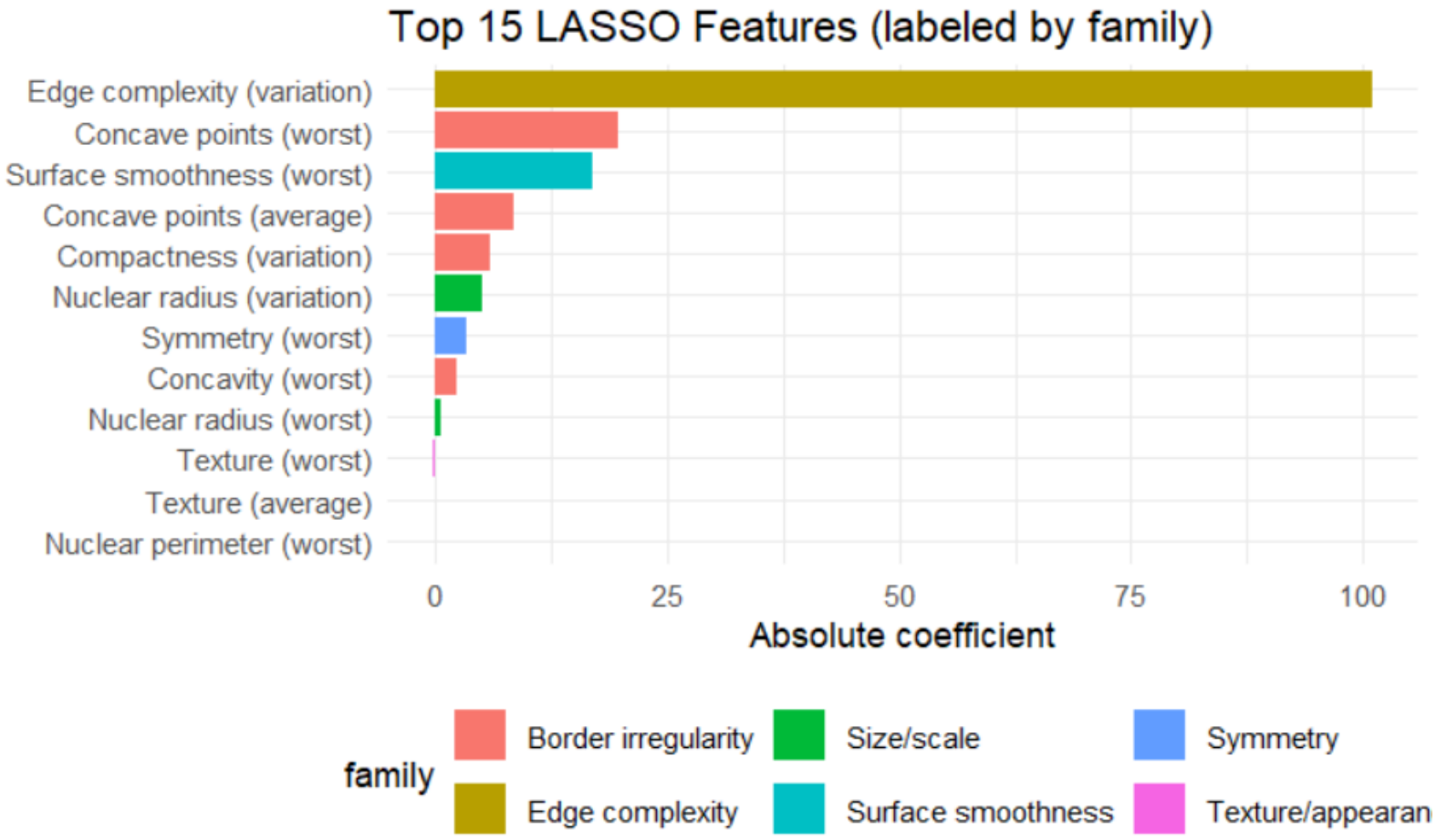
Test AUC = 0.996.

- Accuracy = 0.982
- Sensitivity = 0.952
- Specificity = 1
- Kappa = 0.962
- Balanced Accuracy = 0.976

LASSO INTERPRETATION

Which feature families matter?

- Positive contributors: size/scale and border irregularity
- Negative contributors: surface smoothness, symmetry, edge complexity.
- This matches cytology i.e. **bigger, more irregular, less smooth** nuclei means malignancy.



LDA RESULTS/DISCUSSION

- LDA used to classify tumors
- Benign tumors clustered at lower LD1 values, malignant at higher values
- Clear separation of majority of samples supports predictive value
- Accuracy: 96.5%
- Sensitivity: 92.1%
- Specificity: 99.1%

LDA RESULTS/DISCUSSION

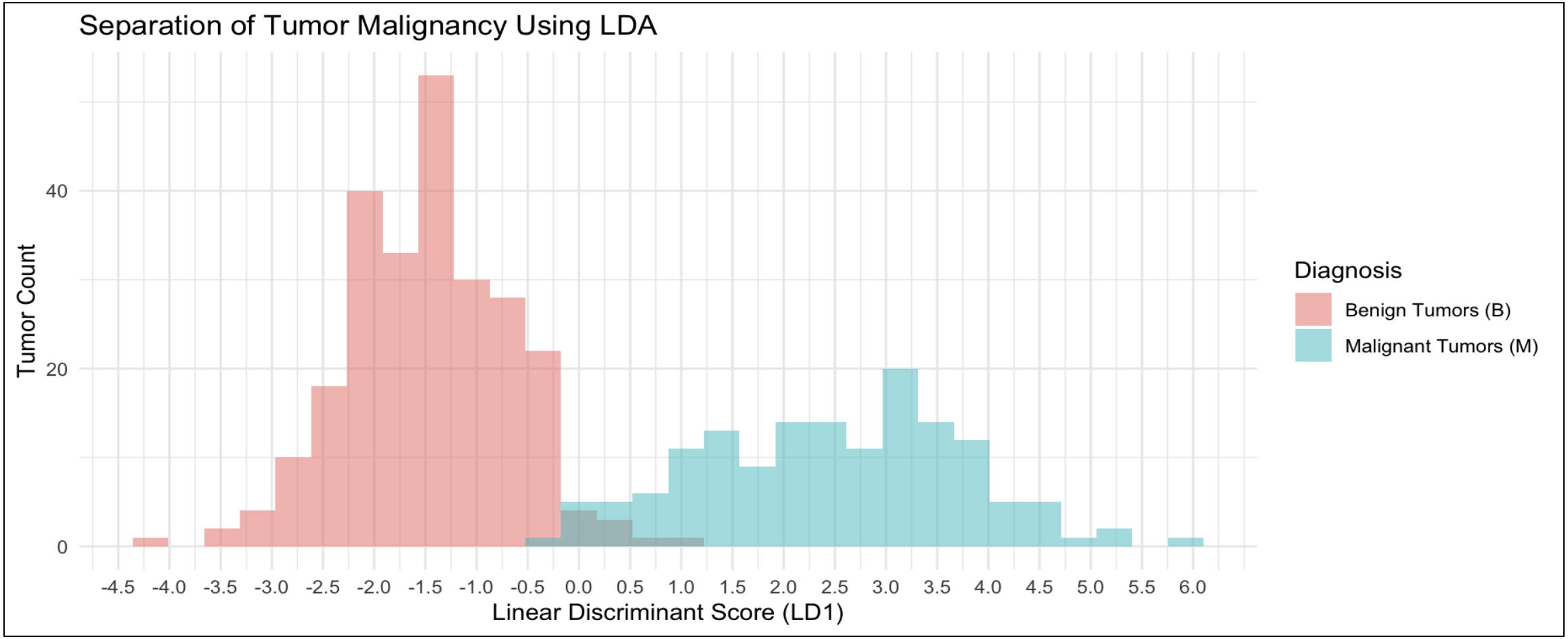


Figure 7 – Histogram distribution along LD1 for benign and malignant tumor data points.

LIMITATIONS

- Controlled public dataset
  - May not capture real-world variability and characteristics
- Only fine-needle aspirate samples used
  - Does not represent all sample types
- Focus on morphological characteristics
  - Excludes important considerations like genetics and clinical history
- Only linear methods applied
  - Nonlinear models could capture other important aspects of the data

FUTURE WORK

- Include more variables in the dataset
  - Demographics, genetics, socioeconomic factors
- Increase dataset size and diversity
  - Larger sample size
  - More data sources
- Explore nonlinear and advanced relationships
- Adapt models to be used in real-world clinical settings

CONCLUSION

- Tumor size features were the strongest predictors of malignancy, followed by shape irregularities
- Findings support FNA as a less invasive, cost-effective screening tools
  - Less reliance on biopsies, fewer unnecessary invasive procedures
- Potential for more accessible early detection of high-risk tumors

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